

onto

$$R_1^2/A \times A \times A \times B,$$

for any numbers A and B satisfying the following conditions.

1. $R_1 < A$.
2. $A^2 < R_2 R_3$.
3. $A^3 B < R_1 R_2 R_3 R_4$.

Construction:

Let I be a quasi-isometric embedding of $R_1 \times A^2/R_1 \times AB/R_1$ into $R_2 \times R_3 \times R_4$.

The conditions 1, 2, and 3 above guarantee that such an embedding exists.

Let U be the union of $R_1 \times \text{image}(I)$ with $\{0\} \times R_2 \times R_3 \times R_4$.

Step 1. There is a Lipshitz retraction of R onto U.

Step 2. There is a retraction of U onto $R_1 \times \text{image}(I)$. This retraction has no bound on its 2-dilation, but it does have the following good property. For each point in U, either the retraction is 2-contracting or else that point is mapped into the plane $x_1 = 0$ and its tangent space is mapped into the tangent space of that plane.

Step 3. There is a quasi-isometric diffeomorphism from $R_1 \times \text{image}(I)$ to $R_1 \times R_1 \times A^2/R_1 \times AB/R_1$, which preserves the coordinate x_1 .

Step 4. There is a degree 1 Lipshitz pinch map from this rectangle onto itself, whose restriction to the plane $x_1 = 0$ is $p(0, x_2, x_3, x_4) = (0, 0, x_3, x_4)$.

Step 5. Compose with the linear map 2-contracting map from this rectangle to $R_1^2/A \times A \times A \times B$. This map preserves the 2-plane $x_1 = 0, x_2 = 0$.

Step 6. There is a Lipshitz pinch map p from this rectangle to itself, whose restriction to the 2-plane $x_1 = 0, x_2 = 0$ is given by $p(0, 0, x_3, x_4) = (0, 0, 0, x_4)$.

The composition of all of these maps is a degree 1 map from $R_1 \times R_2 \times R_3 \times R_4$ to $R_1^2/A \times A \times A \times B$, with 2-dilation less than C.

Final Remarks

In the first part of this chapter, we gave lower bounds for the 2-dilation of diffeomorphisms from R to S. Let $L(R, S)$ be the lower bound for the 2-dilation of any diffeomorphism from R to S which we proved in section 1. In the second section of this chapter, we constructed several non-linear diffeomorphisms with small 2-dilation.